

UML Activity Diagrams

SWE 3633-Software Design &
Architecture

UML Activity Diagrams

- **Activity diagrams** can represent any process and are useful throughout software design.
- A **process** is a collection of related tasks that transforms a set of inputs to a set of outputs.
- **UML activity diagrams** model processes by depicting actions and the flow of control and data between them.

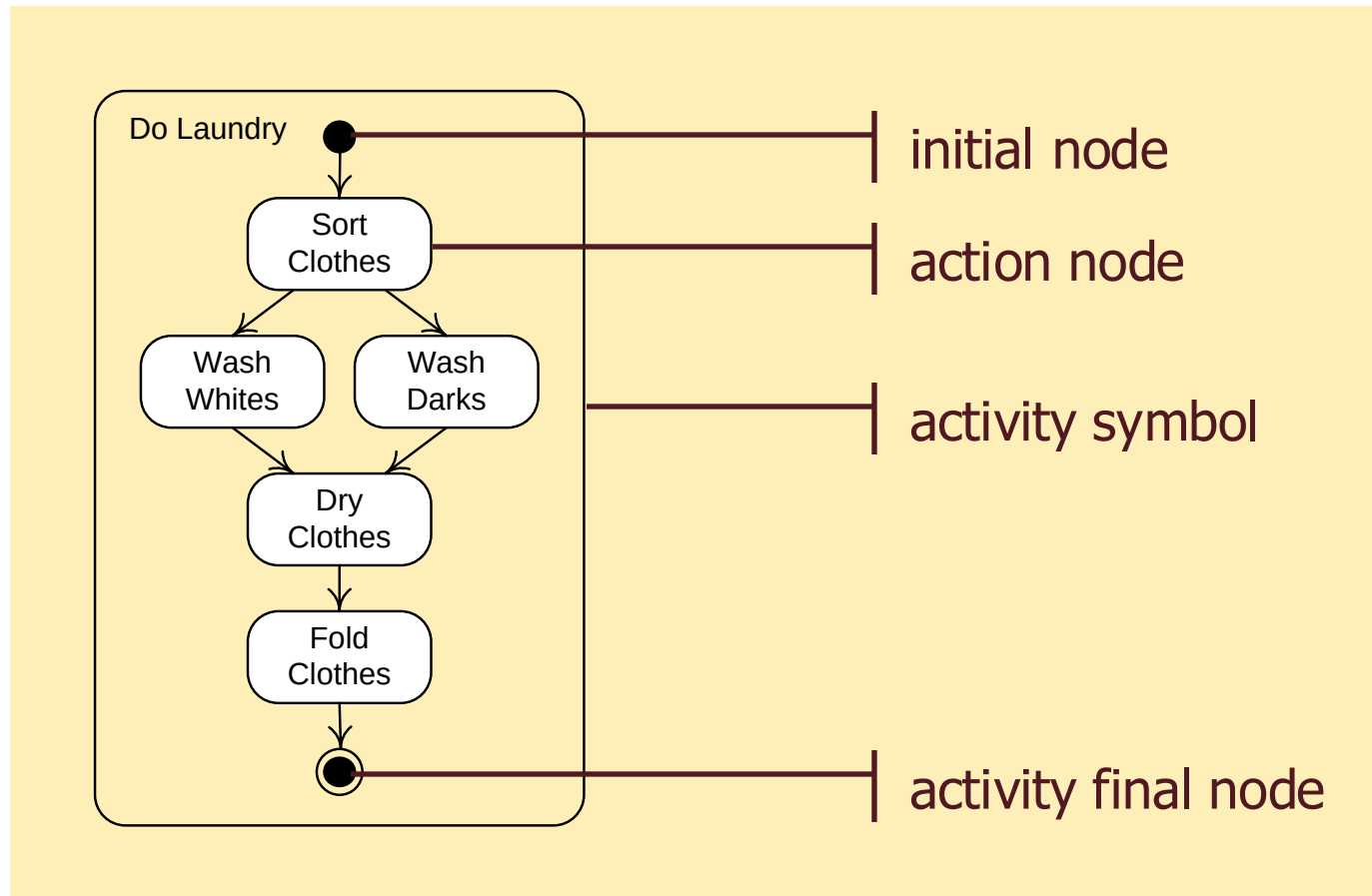
Processes and process descriptions

- A **process** is a collection of related tasks that transforms a set of inputs into a set of outputs.
- Process description notations describe design processes as well as computational processes we design.

Activities and Actions

- An **activity** is a non-atomic task or procedure decomposable into actions.
- An **action** is a task or procedure that cannot be broken into parts.
- A token is a marker used in executing the *virtual machine* specified by an activity diagram.

Activity Graph Elements (page 35)



Execution Model

- **Execution is modeled by tokens that are produced by action nodes, travel over action edges, and are consumed by action nodes.**

Execution Model – cont.

- **When there is a token on every incoming edge of an action node, it consumes them and begins execution.**

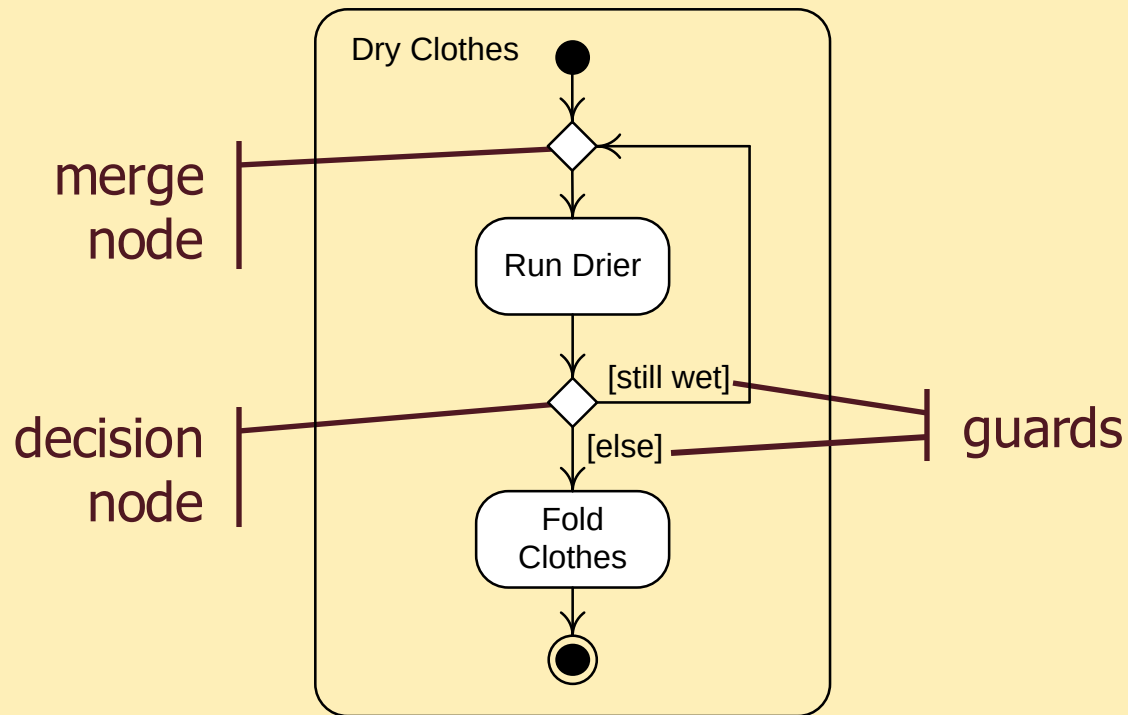
Execution Model – cont.

- **When an action node completes execution, it produces tokens on each of its outgoing edges.**

Execution Model – cont.

- An initial node produces a token on each outgoing edge when an activity begins.
- An activity final node consumes a token available on any incoming edge and terminates the activity.

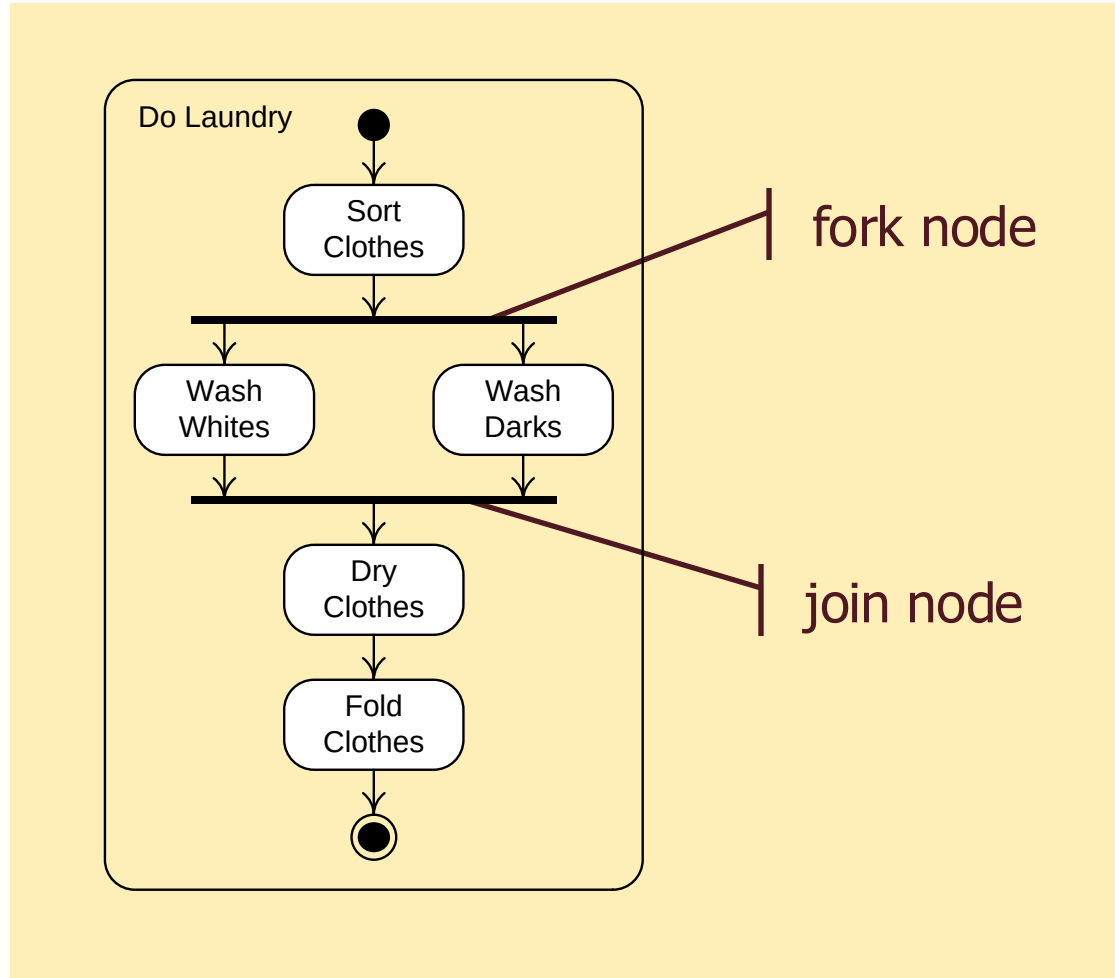
Branching Nodes



Branching Execution

- If a token is made available on the incoming edge of a decision node, the token is made available on the outgoing edge whose guard is true.
- If a token is available on any incoming edge of a merge node, it is made available on its outgoing edge.
- Guards must be mutually exclusive.

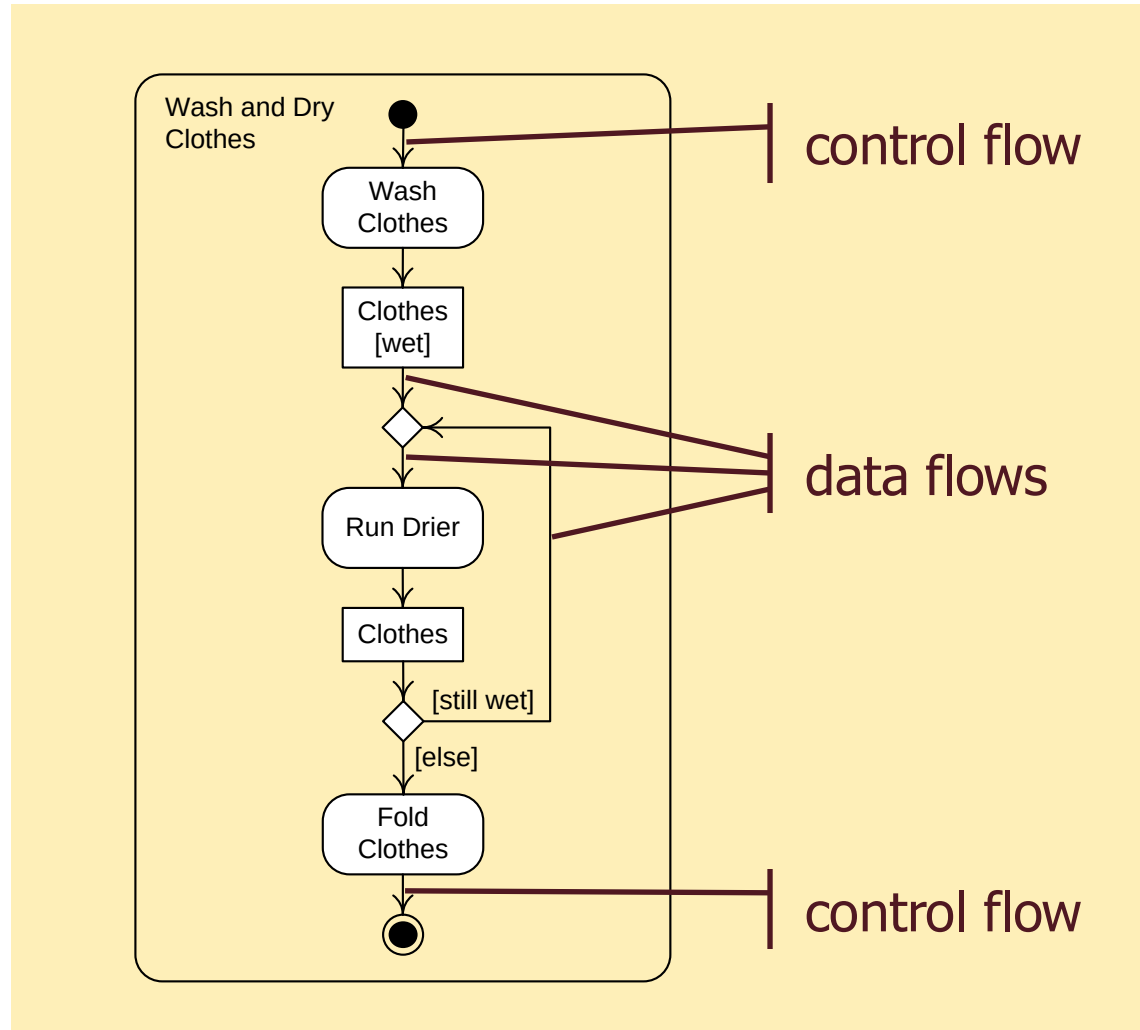
Forking and Joining Nodes



Forking and Joining Execution

- A token available on the incoming edge of a **fork node** is reproduced and made available on all its outgoing edges.
- When tokens are available on every incoming edge of a **join node**, a token is made available on its outgoing edge.
- Concurrency can be modeled without these nodes.

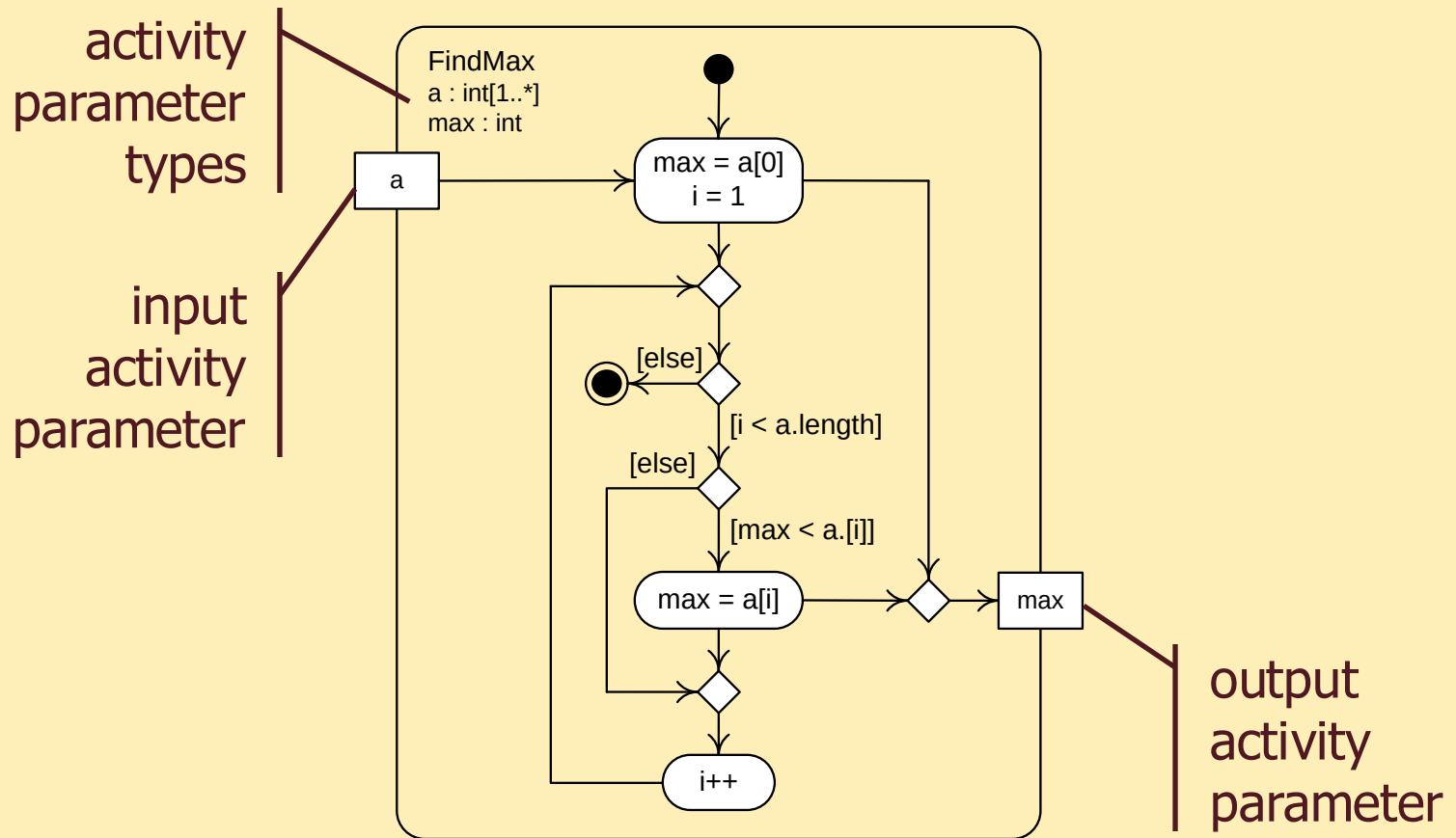
Control and Data Flow Example



Activity Parameters

- **Activity parameters are object nodes placed on activity symbol boundaries to indicate data or object inputs or outputs.**
- **Activity parameters contain the data or object name.**
- **Activity parameter types are specified in the activity symbol beneath the activity name.**

Activity Parameter Example



- **Also see these Figures (for more examples):**
 - **Figure 2-1-6 Concurrent Problem Solution (page 39)**
 - **Figure 2-1-9 Pins and Data Flows (page 42)**

Activity Diagram Heuristics

- Flow control and objects down the page and left to right.
- Name activities and actions with verb phrases.
- Name object nodes with noun phrases.
- Don't use both control and data flows when a data flow alone can do the job.
- Make sure that all nodes entering an action node can provide tokens concurrently.
- Use the [else] guard at every branch.

When to Use Activity Diagrams

When making a dynamic model of any process.

- **During analysis**
- **During resolution**

Questions

